

THE USE OF THE DIFFERENTIAL OPERATORS Δ_Y AND E_Y IN DERIVING SOME IDENTITIES IN ZONAL POLYNOMIALS

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Key words : Sugiura-Fujikoshi coefficients.

Summary : The Laplace-Beltrami differential operator Δ_Y and the Euler differential operator E_Y used by Muirhead can be used to simplify the proofs of existing identities in zonal polynomials attributed to both Muirhead and Sugiura and Fujikoshi as well as to prove further identities of the same type. These new identities can in turn be used to simplify the proofs of other classes of identities.

1. Introduction

The need for algebraic identities involving zonal polynomials first arose when approximations for non-central multivariate distributions were sought. Several researchers have worked on proofs of these identities, most of which are complicated algebraic proofs.

James (1968) proved that zonal polynomials were eigenfunctions of the Laplace-Beltrami operator. Muirhead (1970a) used this fact to formulate a new approach to the definition of zonal polynomials, proofs of which were set out by him (Muirhead (1982)). This definition of zonal polynomials in terms

of the Laplace-Beltrami operator suggests that a conceptually far easier approach to these proofs would be in terms of differential operators.

This paper endeavours to show that not only can the existing identities be proved more easily using the Laplace-Beltrami operator with Euler's differential operator which he first defined in the first half of the eighteenth century (Bell (1937)) but also that the technique leads to the establishment of further new identities which can in turn be used for establishing results involving generalised binomial coefficients and zonal polynomials.

2. Definitions

DEFINITION 2.1

Let Y be an $m \times m$ symmetric matrix with latent roots y_i , $i=1, \dots, m$ and let $\mathbf{k} = (k_1, \dots, k_m)$ be a partition of k into not more than m parts.

Muirhead (1982) defines the zonal polynomial of Y corresponding to $\mathbf{k}$, $C_{\mathbf{k}}(Y)$, to be a symmetric homogeneous polynomial of degree k in the latent roots y_i , $i=1, \dots, m$ such that

$$(i) \quad C_{\mathbf{k}}(Y) = d_{\mathbf{k}} y_1^{k_1} \dots y_m^{k_m} + \text{terms of lower weight,}$$

$d_{\mathbf{k}}$ a constant

$$(ii) \quad C_{\mathbf{k}}(Y) \text{ is an eigenfunction of the differential operator } \Delta_Y,$$

where this Laplace-Beltrami operator is defined to be

$$(2.1) \quad \Delta_Y = \sum_{i=1}^m y_i^2 \frac{\partial^2}{\partial y_i^2} + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} \frac{\partial}{\partial y_i}$$

- (iii) as $\mathbf{k}$ varies over all the partitions of k , the zonal polynomials have unit coefficients in the expansion of $(\text{tr } \mathbf{Y})^k$,

i.e.
$$(\text{tr } \mathbf{Y})^k = (y_1 + \dots + y_m)^k = \sum_{\mathbf{k}} C_{\mathbf{k}}(\mathbf{Y}).$$

DEFINITION 2.2

The Euler differential operator $E_{\mathbf{Y}}$ is defined to be

$$(2.2) \quad E_{\mathbf{Y}} = \sum_{i=1}^m y_i \frac{\partial}{\partial y_i}.$$

LEMMA 2.3

$$(2.3) \quad E_{\mathbf{Y}} \Delta_{\mathbf{Y}} - \Delta_{\mathbf{Y}} E_{\mathbf{Y}} = 0.$$

PROOF

$$\begin{aligned} (i) \quad E_{\mathbf{Y}} \Delta_{\mathbf{Y}} &= \sum_{i=1}^m y_i \frac{\partial}{\partial y_i} \left(\sum_{i=1}^m y_i^2 \frac{\partial^2}{\partial y_i^2} + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} \frac{\partial}{\partial y_i} \right) \\ &= \sum_{i=1}^m y_i \left[2y_i \frac{\partial^2}{\partial y_i^2} + y_i^2 \frac{\partial^3}{\partial y_i^3} + \sum_{\substack{j=1 \\ j \neq i}}^m y_j^2 \frac{\partial^3}{\partial y_i \partial y_j^2} \right. \\ &\quad \left. + \sum_{\substack{j=1 \\ j \neq i}}^m \left(\frac{2y_i(y_i - y_j) - y_i^2}{(y_i - y_j)^2} + \frac{y_i y_j}{(y_i - y_j)^2} \right) \frac{\partial}{\partial y_i} + \sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} \frac{\partial^2}{\partial y_i^2} \right] \end{aligned}$$

$$\begin{aligned}
& + \sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_j^2}{y_j - y_i} \frac{\partial^2}{\partial y_i \partial y_j} + \sum_{\substack{j=1 \\ j \neq i}}^m \sum_{\substack{k=1 \\ k \neq j \\ k \neq i}}^m \frac{y_j^2}{y_j - y_k} \frac{\partial^2}{\partial y_i \partial y_j} \Big] \\
& = \sum_{i=1}^m \left(2y_i^2 \frac{\partial^2}{\partial y_i^2} + y_i^3 \frac{\partial^3}{\partial y_i^3} \right)
\end{aligned}$$

$$\begin{aligned}
& + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \left(y_i y_j^2 \frac{\partial^3}{\partial y_i \partial y_j^2} + \frac{y_i^2}{y_i - y_j} \frac{\partial}{\partial y_i} + \frac{y_i^3}{y_i - y_j} \frac{\partial^2}{\partial y_i^2} \right. \\
& \left. + \frac{y_i y_j^2}{y_j - y_i} \frac{\partial^2}{\partial y_i \partial y_j} \right) + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \sum_{\substack{k=1 \\ k \neq j \\ k \neq i}}^m \frac{y_i y_j^2}{y_j - y_k} \frac{\partial^2}{\partial y_i \partial y_j}
\end{aligned}$$

$$\begin{aligned}
\text{(ii) } \Delta_Y E_Y &= \sum_{i=1}^m y_i^2 \frac{\partial^2}{\partial y_i^2} \left(\sum_{i=1}^m y_i \frac{\partial}{\partial y_i} \right) \\
& + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} \frac{\partial}{\partial y_i} \left(\sum_{i=1}^m y_i \frac{\partial}{\partial y_i} \right) \\
& = \sum_{i=1}^m \left[y_i^2 \left(y_i \frac{\partial^3}{\partial y_i^3} + 2 \frac{\partial^2}{\partial y_i^2} + \sum_{\substack{j=1 \\ j \neq i}}^m y_j \frac{\partial^3}{\partial y_i^2 \partial y_j} \right) \right] \\
& + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} \left(\frac{\partial}{\partial y_i} + y_i \frac{\partial^2}{\partial y_i^2} + y_j \frac{\partial^2}{\partial y_i \partial y_j} \right)
\end{aligned}$$

$$\begin{aligned}
& + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \sum_{\substack{k=1 \\ k \neq j \\ k \neq i}}^m \frac{y_i - y_j^2}{y_j - y_k} \frac{\partial^2}{\partial y_i \partial y_j} \\
& = \sum_{i=1}^m \left(y_i^3 \frac{\partial^3}{\partial y_i^3} + 2y_i^2 \frac{\partial^2}{\partial y_i^2} \right) \\
& \quad + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \left(y_i y_j^2 \frac{\partial^3}{\partial y_i \partial y_j^2} + \frac{y_i^2}{y_i - y_j} \frac{\partial}{\partial y_i} + \frac{y_i^3}{(y_i - y_j)} \frac{\partial^2}{\partial y_i^2} \right. \\
& \quad \left. + \frac{y_i y_j^2}{y_j - y_i} \frac{\partial^2}{\partial y_i \partial y_j} \right) + \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \sum_{\substack{k=1 \\ k \neq j \\ k \neq i}}^m \frac{y_i y_j^2}{y_j - y_k} \frac{\partial^2}{\partial y_i \partial y_j}
\end{aligned}$$

and the result follows.

3. Auxiliary results

The following lemmas will be required for the algebraic simplification in the derivation of the identities.

LEMMA 3.1

$$(3.1) \quad \sum_{i=1}^m \left(\sum_{j=i+1}^m \frac{y_i^2 - y_j^2}{y_i - y_j} - (m-1)y_i \right) = 0.$$

PROOF. This follows directly from factorisation and algebraic simplification.

LEMMA 3.2

$$(3.2) \quad 2 \sum_{i=1}^m \left(\sum_{j=i+1}^m \frac{y_i^3 - y_j^3}{y_i - y_j} - (m-1)y_i^2 \right) = (\operatorname{tr} Y)^2 - \operatorname{tr} Y^2.$$

PROOF.

$$\begin{aligned} & \sum_{i=1}^m \left(\sum_{j=i+1}^m \frac{y_i^3 - y_j^3}{y_i - y_j} - (m-1)y_i^2 \right) \\ &= \sum_{i=1}^m \sum_{j=i+1}^m (y_i^2 + y_i y_j + y_j^2) - \sum_{i=1}^m (m-1)y_i^2 \\ &= \sum_{i=1}^m \sum_{j=i+1}^m y_i y_j = \frac{1}{2} \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m y_i y_j = \frac{1}{2} [(\operatorname{tr} Y)^2 - \operatorname{tr} Y^2]. \end{aligned}$$

LEMMA 3.3

$$(3.3) \quad \sum_{i=1}^m \left(\sum_{j=i+1}^m \frac{y_i^4 - y_j^4}{y_i - y_j} - (m-1)y_i^3 \right) = \operatorname{tr} Y \operatorname{tr} Y^2 - \operatorname{tr} Y^3.$$

LEMMA 3.4

$$(3.4) \quad 2 \sum_{i=1}^m \left(\sum_{j=i+1}^m \frac{y_i^5 - y_j^5}{y_i - y_j} - (m-1)y_i^4 \right) = 2 \operatorname{tr} Y \operatorname{tr} Y^3 + (\operatorname{tr} Y^2)^2 - 3 \operatorname{tr} Y^4.$$

The proofs of lemma 3.3 and lemma 3.4 follow the same pattern as the proof of lemma 3.2.

LEMMA 3.5

$$\sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^k}{y_i - y_j} \right) = \sum_{i=1}^m \sum_{j=i+1}^m \frac{y_i^k - y_j^k}{y_i - y_j} \quad k = 2, 3, 4, 5.$$

PROOF. This can be proved by mathematical induction for each value of k .

4. Application of E_Y and Δ_Y to $C_k(Y)$

LEMMA 4.1.

$$(4.1) \quad E_Y C_k(Y) = k C_k(Y)$$

i.e. $C_k(Y)$ is an eigenfunction of E_Y with eigenvalue k .

PROOF. This result was established by James (1968) and follows from definition 2.1 (i) since $C_k(Y)$ is a homogeneous polynomial of degree k .

LEMMA 4.2

$$(4.2) \quad \Delta_Y C_k(Y) = [a_1(k) + k(m-1)] C_k(Y).$$

PROOF. This was proved by James (1968) and the proof is also given in Muirhead (1982) where $a_1(k)$ is one of the Sugiura-Fujikoshi (1969) coefficients which are defined as follows:

$$a_1(\mathbf{k}) = \sum_{i=1}^m k_i(k_i - i)$$

$$a_2(\mathbf{k}) = \sum_{i=1}^m k_i(4k_i^2 - 6ik_i + 3i^2)$$

$$a_3(\mathbf{k}) = \sum_{i=1}^m k_i(2k_i^3 - 4ik_i^2 + 3i^2k_i - i^3).$$

LEMMA 4.3

$$(4.3) \quad T_Y C_{\mathbf{k}}(Y) = (\Delta_Y - (m-1)E_Y) C_{\mathbf{k}}(Y) = a_1(\mathbf{k}) C_{\mathbf{k}}(Y)$$

i.e. $C_{\mathbf{k}}(Y)$ is an eigenfunction of $T_Y = (\Delta_Y - (m-1)E_Y)$, say, with eigenvalue $a_1(\mathbf{k})$.

PROOF. This follows on combining (4.1) and (4.2) and was established in Muirhead (1982).

5. Identities

5.1 Existing identities

Identity 5.1.1

$$(5.1.1) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{C_{\mathbf{k}}(Y)}{k!} = \text{etr } Y.$$

PROOF. This follows from definition 2.1(iii) and is given in Muirhead (1982)

Identity 5.1.2

$$(5.1.2) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{k C_{\mathbf{k}}(\mathbf{Y})}{k!} = \text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}.$$

PROOF. Applying $E_{\mathbf{Y}}$ to (5.1.1), as proved by Muirhead (1982), gives

$$\begin{aligned} E_{\mathbf{Y}} \left(\sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{C_{\mathbf{k}}(\mathbf{Y})}{k!} \right) &= \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{E_{\mathbf{Y}} C_{\mathbf{k}}(\mathbf{Y})}{k!} \\ &= \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{k C_{\mathbf{k}}(\mathbf{Y})}{k!} \quad \text{by (4.1)} \end{aligned}$$

$$E_{\mathbf{Y}} \text{etr } \mathbf{Y} = \sum_{i=1}^m y_i \frac{\partial}{\partial y_i} (\text{etr } \mathbf{Y})$$

$$= \sum_{i=1}^m y_i \text{etr } \mathbf{Y}$$

$$= \text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}.$$

and the result follows.

Subramaniam (1976) extended this result to

Identity 5.1.3

$$(5.1.3) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{C_{\mathbf{k}}(\mathbf{Y})}{(k-r)!} = \text{etr } \mathbf{Y} (\text{tr } \mathbf{Y})^r.$$

This identity is included for the sake of completeness as it is frequently used, for different values of r , in the application of the identities.

Identity 5.1.4

$$(5.1.4) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_{\mathbf{I}}(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}^2.$$

PROOF. Suggested by Muirhead (1982).

Application of $T_{\mathbf{Y}}$ to LHS of (5.1.1) gives LHS of (5.1.4), using (4.3).

Applying $T_{\mathbf{Y}}$ to RHS of (5.1.1) we get that, since

$$\begin{aligned} T_{\mathbf{Y}} &= \sum_{i=1}^m y_i^2 \frac{\partial^2}{\partial y_i^2} + \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1)y_i \right) \frac{\partial}{\partial y_i}, \\ T_{\mathbf{Y}} \text{etr } \mathbf{Y} &= \sum_{i=1}^m y_i^2 \frac{\partial^2}{\partial y_i^2} \text{etr } \mathbf{Y} + \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1)y_i \right) \frac{\partial}{\partial y_i} \text{etr } \mathbf{Y} \\ &= \text{etr } \mathbf{Y} \sum_{i=1}^m y_i^2 + \text{etr } \mathbf{Y} \left[\sum_{i=1}^m \left(\sum_{\substack{j=i+1}}^m \frac{y_i^2 - y_j^2}{y_i - y_j} - (m-1)y_i \right) \right] \\ &= \text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}^2 \quad \text{using lemma 3.1.} \end{aligned}$$

Identity 5.1.5

$$(5.1.5) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{(k-2)a_{\mathbf{I}}(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \text{etr } \mathbf{Y} \text{ tr } \mathbf{Y} \text{ tr } \mathbf{Y}^2.$$

PROOF. Application of $E_{\mathbf{Y}}$ to LHS of (5.1.4) gives, using (4.1),

$$\sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{k a_{\mathbf{k}}(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!}.$$

Applying $E_{\mathbf{Y}}$ to RHS of (5.1.4) we get

$$\text{etr } \mathbf{Y} \text{ tr } \mathbf{Y} \text{ tr } \mathbf{Y}^2 + 2 \text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}^2,$$

using the same technique as was used in the proof of identity 5.1.2. Combining this result with identity 5.1.4 gives identity 5.1.5. Note that identity 5.1.5 can also be obtained by applying $T_{\mathbf{Y}}$ to (5.1.2) and using lemma 3.1. This serves as an illustration of lemma 2.3.

Identity 5.1.6

$$(5.1.6) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_{\mathbf{k}}^2(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \text{etr } \mathbf{Y} [(tr \mathbf{Y})^2 + tr \mathbf{Y}^2 + 4 tr \mathbf{Y}^3 + (tr \mathbf{Y}^2)^2].$$

PROOF. This identity was first proved by Sugiura and Fujikoshi in 1969 using algebraic expansions. The result can be obtained more directly by applying $T_{\mathbf{Y}}$ to identity 5.1.4 and using lemma 3.2 as follows:

$$T_{\mathbf{Y}} \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_{\mathbf{k}}(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_{\mathbf{k}}^2(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!}$$

$$T_{\mathbf{Y}} (\text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}^2) = \sum_{i=1}^m y_i^2 \frac{\partial^2}{\partial y_i^2} (\text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}^2)$$

$$+ \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1)y_i \right) \frac{\partial}{\partial y_i} (\text{etr } \mathbf{Y} \text{ tr } \mathbf{Y}^2)$$

$$\begin{aligned}
&= \sum_{i=1}^m y_i^2 \frac{\partial}{\partial y_i} [\text{etr } Y \text{ tr } Y^2 + 2 \text{etr } Y y_i] \\
&\quad + \text{etr } Y \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1)y_i \right) (\text{tr } Y^2 + 2y_i) \\
&= \sum_{i=1}^m y_i^2 [\text{etr } Y \text{ tr } Y^2 + 4 \text{etr } Y y_i + 2 \text{etr } Y] \\
&\quad + \text{etr } Y \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1)y_i \right) (\text{tr } Y^2) \\
&\quad + 2 \text{etr } Y \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^3}{y_i - y_j} - (m-1)y_i^2 \right) \\
&= \text{etr } Y \{ (\text{tr } Y^2)^2 + 4 \text{tr } Y^3 + 2 \text{tr } Y^2 \} \\
&\quad + \text{etr } Y \cdot 0 \quad \text{using lemma 3.1} \\
&\quad + 2 \text{etr } Y \sum_{i=1}^m \left(\sum_{j=i+1}^m \frac{y_i^3 - y_j^3}{y_i - y_j} - (m-1)y_i^2 \right) \\
&= \text{etr } Y \left[(\text{tr } Y^2)^2 + 4 \text{tr } Y^3 + 2 \text{tr } Y^2 + 2 \sum_{i=1}^m \sum_{\substack{j=1 \\ j > i}}^m y_i y_j \right] \\
&= \text{etr } Y \left[(\text{tr } Y^2)^2 + 4 \text{tr } Y^3 + 2 \text{tr } Y^2 + [(\text{tr } Y)^2 - \text{tr } (Y^2)] \right] \\
&\quad \text{using lemma 3.2} \\
&= \text{etr } Y [(\text{tr } (Y^2))^2 + 4 \text{tr } Y^3 + \text{tr } Y^2 + (\text{tr } Y)^2].
\end{aligned}$$

5.2 New identities

Identity 5.2.1

$$(5.2.1) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{k a_1^2(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \text{etr } \mathbf{Y} [2 (\text{tr } \mathbf{Y})^2 + (\text{tr } \mathbf{Y})^3 + \text{tr } \mathbf{Y} \text{tr } \mathbf{Y}^2 \\ + 2 \text{tr } \mathbf{Y}^2 + 4 (\text{tr } \mathbf{Y}^2)^2 + 4 \text{tr } \mathbf{Y} \text{tr } \mathbf{Y}^3 \\ + 12 \text{tr } \mathbf{Y}^3 + \text{tr } \mathbf{Y} (\text{tr } \mathbf{Y}^2)^2].$$

PROOF. Application of $E_{\mathbf{Y}}$ to identity 5.1.6 gives

$$E_{\mathbf{Y}} \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_1^2(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_1^2(\mathbf{k}) E_{\mathbf{Y}} C_{\mathbf{k}}(\mathbf{Y})}{k!} = \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{k a_1^2(\mathbf{k})}{k!} C_{\mathbf{k}}(\mathbf{Y})$$

$$\begin{aligned} & E_{\mathbf{Y}} [\text{etr } \mathbf{Y} ((\text{tr } \mathbf{Y})^2 + \text{tr } \mathbf{Y}^2 + 4 \text{tr } \mathbf{Y}^3 + (\text{tr } \mathbf{Y}^2)^2)] \\ &= \sum_{i=1}^m y_i \frac{\partial}{\partial y_i} \{ \text{etr } \mathbf{Y} [(\text{tr } \mathbf{Y})^2 + \text{tr } \mathbf{Y}^2 + 4 \text{tr } \mathbf{Y}^3 \\ &\quad + (\text{tr } \mathbf{Y}^2)^2] \} \\ &= \sum_{i=1}^m y_i \{ \text{etr } \mathbf{Y} [(\text{tr } \mathbf{Y})^2 + 2 \text{tr } \mathbf{Y} + \text{tr } \mathbf{Y}^2 + 2y_i \\ &\quad + 4 \text{tr } \mathbf{Y}^3 + 12y_i^2 + (\text{tr } \mathbf{Y}^2)^2 + 4 \text{tr } \mathbf{Y}^2 \cdot y_i] \} \\ &= \text{etr } \mathbf{Y} [(\text{tr } \mathbf{Y})^3 + 2 (\text{tr } \mathbf{Y})^2 + \text{tr } \mathbf{Y} \text{tr } \mathbf{Y}^2 \\ &\quad + 2 \text{tr } \mathbf{Y}^2 + 4 \text{tr } \mathbf{Y} \text{tr } \mathbf{Y}^3 + 12 \text{tr } \mathbf{Y}^3 \\ &\quad + \text{tr } \mathbf{Y} (\text{tr } \mathbf{Y}^2)^2 + 4 (\text{tr } \mathbf{Y}^2)^2] \end{aligned}$$

Identity 5.2.2

$$(5.2.2) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_{\mathbf{k}}^3(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \text{etr } \mathbf{Y} [3 (\text{tr } \mathbf{Y})^2 \text{tr } \mathbf{Y}^2 + 16 \text{tr } \mathbf{Y} \text{tr } \mathbf{Y}^2 \\ + 3 \text{tr } \mathbf{Y}^2 + 3 (\text{tr } \mathbf{Y}^2)^2 + 16 \text{tr } \mathbf{Y}^3 + (\text{tr } \mathbf{Y})^2 \\ + 12 \text{tr } \mathbf{Y}^2 \text{tr } \mathbf{Y}^3 + 32 \text{tr } \mathbf{Y}^4 + (\text{tr } \mathbf{Y}^2)^3].$$

PROOF. Application of $T_{\mathbf{Y}}$ to identity 5.1.6 gives

$$T_{\mathbf{Y}} \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_{\mathbf{k}}^2(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} = \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_{\mathbf{k}}^3(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} \quad \text{by lemma 4.3}$$

$$T_{\mathbf{Y}} \{ \text{etr } \mathbf{Y} [(\text{tr } \mathbf{Y})^2 + \text{tr } \mathbf{Y}^2 + 4 \text{tr } \mathbf{Y}^3 + (\text{tr } \mathbf{Y}^2)^2] \}$$

$$= \sum_{i=1}^m y_i^2 \frac{\partial}{\partial y_i} \{ \text{etr } \mathbf{Y} [(\text{tr } \mathbf{Y})^2 + \text{tr } \mathbf{Y}^2 + 4 \text{tr } \mathbf{Y}^3 + (\text{tr } \mathbf{Y}^2)^2] \}$$

$$+ \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1)y_i \right) \frac{\partial}{\partial y_i} \{ \text{etr } \mathbf{Y} [(\text{tr } \mathbf{Y})^2 + \text{tr } \mathbf{Y}^2 \\ + 4 \text{tr } \mathbf{Y}^3 + (\text{tr } \mathbf{Y}^2)^2] \}$$

$$= \sum_{i=1}^m y_i^2 \frac{\partial}{\partial y_i} \{ \text{etr } \mathbf{Y} [(\text{tr } \mathbf{Y})^2 + 2 \text{tr } \mathbf{Y} + \text{tr } \mathbf{Y}^2 + 2y_i \\ + 4 \text{tr } \mathbf{Y}^3 + 12y_i^2 + (\text{tr } \mathbf{Y}^2)^2 + 4 \text{tr } \mathbf{Y}^2 y_i] \}$$

$$+ \text{etr } \mathbf{Y} \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1)y_i \right) \{ (\text{tr } \mathbf{Y})^2$$

$$\begin{aligned}
& + 2 \operatorname{tr} \mathbf{Y} + \operatorname{tr} \mathbf{Y}^2 + 4 \operatorname{tr} \mathbf{Y}^3 + (\operatorname{tr} \mathbf{Y}^2)^2 + (2 + 4 \operatorname{tr} \mathbf{Y}^2) y_i \\
& + 12 y_i^2 \} \\
= & \sum_{i=1}^m y_i^2 \operatorname{etr} \mathbf{Y} [(\operatorname{tr} \mathbf{Y})^2 + 4 \operatorname{tr} \mathbf{Y} + 4 + 5 \operatorname{tr} \mathbf{Y}^2 + 4 y_i + 4 \operatorname{tr} \mathbf{Y}^3 \\
& + 32 y_i^2 + 24 y_i + (\operatorname{tr} \mathbf{Y}^2)^2 + 8 \operatorname{tr} \mathbf{Y}^2 y_i] \\
& + \operatorname{etr} \mathbf{Y} \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^2}{y_i - y_j} - (m-1) y_i \right) \{ (\operatorname{tr} \mathbf{Y})^2 \\
& + 2 \operatorname{tr} \mathbf{Y} + \operatorname{tr} \mathbf{Y}^2 + 4 \operatorname{tr} \mathbf{Y}^3 + (\operatorname{tr} \mathbf{Y}^2)^2 \} \\
& + \operatorname{etr} \mathbf{Y} \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^3}{y_i - y_j} - (m-1) y_i^2 \right) (2 + 4 \operatorname{tr} \mathbf{Y}^2) \\
& + 12 \operatorname{etr} \mathbf{Y} \sum_{i=1}^m \left(\sum_{\substack{j=1 \\ j \neq i}}^m \frac{y_i^4}{y_i - y_j} - (m-1) y_i^3 \right) \\
= & \operatorname{etr} \mathbf{Y} \{ (\operatorname{tr} \mathbf{Y})^2 \operatorname{tr} \mathbf{Y}^2 + 4 \operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^2 + 4 \operatorname{tr} \mathbf{Y}^2 + 5 (\operatorname{tr} \mathbf{Y}^2)^2 \\
& + 28 \operatorname{tr} \mathbf{Y}^3 + 4 \operatorname{tr} \mathbf{Y}^3 \operatorname{tr} \mathbf{Y}^2 + 32 \operatorname{tr} \mathbf{Y}^4 + (\operatorname{tr} \mathbf{Y}^2)^3 \\
& + 8 \operatorname{tr} \mathbf{Y}^2 \operatorname{tr} \mathbf{Y}^3 \} + \operatorname{etr} \mathbf{Y} \cdot \mathbf{O} \quad \text{by lemma 3.} \\
& + \operatorname{etr} \mathbf{Y} (1 + 2 \operatorname{tr} \mathbf{Y}^2) ((\operatorname{tr} \mathbf{Y})^2 - \operatorname{tr} \mathbf{Y}^2) \quad \text{by lemma 3.} \\
& + 12 \operatorname{etr} \mathbf{Y} (\operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^2 - \operatorname{tr} \mathbf{Y}^3) \quad \text{by lemma 3.}
\end{aligned}$$

and the result follows.

Identity 5.2.3

$$\begin{aligned}
 (5.2.3) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{k a_1^3(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} &= e \operatorname{tr} \mathbf{Y} [3 (\operatorname{tr} \mathbf{Y})^3 \operatorname{tr} \mathbf{Y}^2 + 28 (\operatorname{tr} \mathbf{Y})^2 \operatorname{tr} \mathbf{Y}^2 \\
 &+ 51 \operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^2 + 3 \operatorname{tr} \mathbf{Y} (\operatorname{tr} \mathbf{Y}^2)^2 + (\operatorname{tr} \mathbf{Y})^3 \\
 &+ 60 \operatorname{tr} \mathbf{Y}^2 \operatorname{tr} \mathbf{Y}^3 + 16 \operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^3 + 6 \operatorname{tr} \mathbf{Y}^2 \\
 &+ 32 \operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^4 + \operatorname{tr} \mathbf{Y} (\operatorname{tr} \mathbf{Y}^2)^3 + 2 (\operatorname{tr} \mathbf{Y})^2 \\
 &+ 12 \operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^2 \operatorname{tr} \mathbf{Y}^3 + 12 (\operatorname{tr} \mathbf{Y}^2)^2 \\
 &+ 48 \operatorname{tr} \mathbf{Y}^3 + 128 \operatorname{tr} \mathbf{Y}^4 + 6 (\operatorname{tr} \mathbf{Y}^2)^3].
 \end{aligned}$$

PROOF. The proof of this identity follows the same lines as the proof of identity 5.2.1. The operator $E_{\mathbf{Y}}$ is applied to both sides of identity 5.2.2 and the result is simplified using lemmas 3.1, 3.2 and 3.3.

Identity 5.2.4

$$\begin{aligned}
 (5.2.4) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{k a_1^4(\mathbf{k}) C_{\mathbf{k}}(\mathbf{Y})}{k!} &= e \operatorname{tr} \mathbf{Y} [6 (\operatorname{tr} \mathbf{Y})^2 (\operatorname{tr} \mathbf{Y}^2)^2 + 48 (\operatorname{tr} \mathbf{Y}^3)^2 \\
 &+ 24 (\operatorname{tr} \mathbf{Y}^2)^2 \operatorname{tr} \mathbf{Y}^3 + 64 (\operatorname{tr} \mathbf{Y}^2)^2 \operatorname{tr} \mathbf{Y} \\
 &+ 88 \operatorname{tr} \mathbf{Y}^2 \operatorname{tr} \mathbf{Y}^3 + 24 (\operatorname{tr} \mathbf{Y})^2 \operatorname{tr} \mathbf{Y}^3 \\
 &+ 128 \operatorname{tr} \mathbf{Y}^2 \operatorname{tr} \mathbf{Y}^4 + 10 (\operatorname{tr} \mathbf{Y})^2 \operatorname{tr} \mathbf{Y}^2 \\
 &+ 216 \operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^3 + 5 \operatorname{tr} \mathbf{Y}^2 + 3 (\operatorname{tr} \mathbf{Y})^4 \\
 &+ 68 \operatorname{tr} \mathbf{Y} \operatorname{tr} \mathbf{Y}^2 + 111 (\operatorname{tr} \mathbf{Y}^2)^2 + 6 (\operatorname{tr} \mathbf{Y}^2)^3 \\
 &+ 124 \operatorname{tr} \mathbf{Y}^3 + 312 \operatorname{tr} \mathbf{Y}^4 + 400 \operatorname{tr} \mathbf{Y}^5 \\
 &+ (\operatorname{tr} \mathbf{Y}^2)^4 + 16 (\operatorname{tr} \mathbf{Y})^3 + 3 (\operatorname{tr} \mathbf{Y})^2].
 \end{aligned}$$

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PROOF. As with the proof of identity 5.2.2, this identity is obtained by applying the operator T_Y , this time to identity 5.2.2, and using lemmas 3.1, 3.2, 3.3 and 3.4.

6. Applications

One of the applications of these new identities is that expressions are available for $\sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_i(\mathbf{k}) C_{\mathbf{k}}(Y)}{k!}$, $\text{etr } Y$ and $(\text{tr } Y^s)^t$ where s, t are positive integers (Sugiyama (1969)), it is now possible, using identity 5.2.2, to obtain an expression for the case $i = 3$, i.e.

$$(6.1) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_3(\mathbf{k}) C_{\mathbf{k}}(Y)}{k!} = \text{etr } Y [2 \text{tr } Y^4 + 4 \text{tr } Y^2 + 4 \text{tr } Y \text{tr } Y^2 +$$

PROOF. Application of the operator T_Y to the following identity, established by Sugiyama and Fujikoshi (1969),

$$\sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_2(\mathbf{k}) C_{\mathbf{k}}(Y)}{k!} = \text{etr } Y [\text{tr } Y + 3(\text{tr } Y)^2 +$$

gives, on algebraic simplification and the use of lemmas 3.1, 3.2 and 3.3,

$$(6.2) \quad \sum_{k=0}^{\infty} \sum_{\mathbf{k}} \frac{a_1(\mathbf{k}) a_2(\mathbf{k}) C_{\mathbf{k}}(Y)}{k!} = \text{etr } Y [25 \text{tr } Y \text{tr } Y^2 + 11 \text{tr } Y^2 + 3(\text{tr } Y)^2 \text{tr } Y^2 + 3(\text{tr } Y^2)^2 + 24 \text{tr } Y^4 + 24 \text{tr } Y^3 + 4 \text{tr } Y^3 \text{tr } Y^2 + 3(\text{tr } Y)^2].$$

Substitution of (6.2) and identities 5.2.2 and 5.1.5 into the following result by Fujikoshi (1970)

$$\begin{aligned} \sum_{k=0}^{\infty} \sum_k \frac{C_k(Y)}{k!} \{ a_1^3(k) - a_1(k)[a_2(k) - k + 2] + 2a_3(k) \} \\ = \text{etr } Y [12 \text{tr } Y^4 + 8 \text{tr } Y^2 \text{tr } Y^3 + (\text{tr } Y^2)^3] \end{aligned}$$

gives (6.1) after further algebraic simplification.

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